
THE DIGITAL COUCH: SIMULATING CROSS-CULTURAL VIEWS ON HAPPINESS WITH LLM PERSONAS

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ABSTRACT

Ensuring Large Language Models (LLMs) are aligned with diverse cultural values is critical for their global adoption, yet modeling abstract concepts like “happiness” remains a significant challenge. This paper introduces a novel methodology to evaluate cultural alignment by simulating a landmark cross-cultural psychology study using detailed LLM personas for the United States and Japan. We generate qualitative data on happiness and unhappiness and use Multidimensional Scaling (MDS) to visualize the underlying conceptual structures. Our results demonstrate that LLM personas produce culturally-consistent themes: US personas associate happiness with personal achievement, while Japanese personas link it to social harmony. Furthermore, we show that a “culturally-aware” system prompt significantly sharpens these cultural distinctions, highlighting contextual priming as an effective technique for eliciting nuanced cultural representations from LLMs.

1 INTRODUCTION

The global deployment of Large Language Models (LLMs) requires their alignment with diverse cultural values to ensure effective and equitable interaction. Models that fail to grasp subtle, culturally-specific concepts risk perpetuating biases and creating poor user experiences. Addressing this challenge, this paper introduces a novel methodology to evaluate and enhance the cultural alignment of LLMs.

The central challenge lies in modeling abstract concepts like “happiness,” which lack universal definitions and are shaped by cultural narratives. In individualistic cultures, happiness is often tied to personal achievement, while in collectivistic cultures, it is more associated with social harmony. Eliciting these nuanced distinctions from an LLM is a complex task requiring methods that go beyond surface-level knowledge.

Our approach simulates a landmark cross-cultural psychology study using detailed LLM personas instead of human participants. We create personas for individuals in Japan and the United States, designed to embody specific cultural and demographic contexts. This allows us to generate rich, qualitative data reflecting distinct cultural perspectives on happiness and unhappiness.

We analyze the qualitative responses using Multidimensional Scaling (MDS) to visualize the conceptual structures of happiness and unhappiness for each cultural group. Our results show that LLM personas generate culturally-consistent themes: US personas associate happiness with personal achievement, while Japanese personas link it to social harmony. Crucially, we find that these cultural distinctions are significantly sharpened by a “culturally-aware” system prompt, demonstrating that explicit contextual priming is a powerful technique for eliciting deeper cultural nuances from LLMs.

Our primary contributions are as follows:

- A methodology for evaluating LLM cultural alignment by simulating a cross-cultural psychology study with detailed personas.
- Empirical evidence that LLM personas can generate culturally-consistent themes for abstract concepts like happiness in Japanese and US contexts.
- A demonstration that a “culturally-aware” system prompt significantly enhances the cultural specificity of LLM responses.

2 RELATED WORK

2.1 EVALUATING CULTURAL ALIGNMENT IN LLMs

individualistic and collectivistic cultures. In individualistic societies like the United States, happiness is often linked to personal achievement and autonomy. In contrast, collectivistic societies like Japan tend to associate happiness with social harmony and interpersonal relationships. Our study investigates whether LLMs can replicate these distinct cultural patterns.

3.2 LLM PERSONAS FOR SIMULATION

Modern LLMs can adopt detailed personas to simulate human responses based on specified backgrounds. By providing an LLM with a rich description of a fictional individual, it can generate responses grounded in that simulated perspective, offering a scalable method for exploring complex social phenomena. We leverage this technique to simulate participants from Japan and the United States, generating culturally-specific qualitative data without the logistical challenges of a large-scale international survey.

3.3 MULTIDIMENSIONAL SCALING

Multidimensional Scaling (MDS) is a statistical technique used to visualize the similarity between items in a dataset. Given a matrix of pairwise dissimilarities, MDS projects items into a low-dimensional space (typically 2D) where the distances between points represent their original dissimilarities—similar items are placed closer together. This method is ideal for revealing the underlying conceptual structure of qualitative data, such as the text from our survey, without requiring predefined categories. We use MDS to map the conceptual landscapes of happiness and unhappiness generated by our Japanese and US personas.

4 METHOD

Our methodology simulates a cross-cultural psychology study in three stages: persona definition, survey administration, and data analysis via Multidimensional Scaling (MDS).

4.1 PERSONA DEFINITION

We defined a set of detailed personas representing individuals from Japan and the United States, each with a profile including demographics, interests, and a life summary. This contextual grounding is crucial for eliciting responses consistent with a specific cultural viewpoint, rather than generic text. For instance, our personas ranged from a Japanese high school student facing university exams to a middle-aged American mother balancing work and family.

4.2 SURVEY ADMINISTRATION

We prompted each persona to provide five distinct examples of “happiness” and “unhappiness.” Our experiment utilized two conditions for generating these responses:

1. **Baseline Prompt:** A standard, neutral system instruction was used.
2. **Culturally-Aware Prompt:** The system instruction was modified to explicitly direct the LLM to consider the persona’s cultural background, values, and social context.

This was designed to test whether enhanced contextual priming elicits more nuanced responses. The resulting qualitative text from both conditions formed our primary dataset.

4.3 DATA ANALYSIS WITH MULTIDIMENSIONAL SCALING

We analyzed the qualitative data using Multidimensional Scaling (MDS). For each response set (e.g., US-Happiness, JP-Happiness), we first constructed a pairwise similarity matrix where an LLM judged the conceptual similarity between every pair of responses. We then applied MDS to this matrix to project the data into a two-dimensional conceptual map. On this map, each point is a single response, and the distance between points is inversely proportional to their conceptual similarity, allowing us to identify underlying thematic clusters.

5 EXPERIMENTAL SETUP

All experiments were orchestrated via our `experiment_py` script. To generate persona responses, we used a large language model with deterministic settings (temperature of 0), a maximum output of 3000 tokens, and a structured JSON response format. Our dataset was generated from five US and four Japanese personas, designed to capture a range of ages and occupations. Each was prompted to produce five examples for both “happiness” and “unhappiness” under two system prompt conditions: a neutral baseline and a “culturally-aware” prompt that explicitly directed the model to consider cultural context.

The resulting text was processed into four datasets (US-Happiness, US-Unhappiness, JP-Happiness, and JP-Unhappiness). For each, a pairwise similarity matrix was generated by using a separate LLM call to score the conceptual similarity between every pair of responses. This matrix then served as input for a standard Multidimensional Scaling (MDS) algorithm, which projected the data into a two-dimensional visualization of the conceptual structure for each cultural group.

6 RESULTS

Our analysis reveals distinct cultural patterns in the conceptualization of happiness and unhappiness. We present our findings in two stages: the baseline condition and the results from using a culturally-aware system prompt.

6.1 BASELINE CONDITION

Under the baseline condition, the MDS plots for happiness (Figure 1) show a clear divergence between cultural groups. The US persona responses cluster around themes of personal achievement and autonomy (e.g., “hitting a goal, like finally solving a tough LeetCode problem”), whereas the Japanese persona responses cluster around social harmony and interpersonal relationships (e.g., “when my family praises my cooking”). These tendencies align with established theories of individualistic and collectivistic cultures. Similarly, the plots for unhappiness (Figure 2) show US personas associating it with personal failure and constraints on freedom, while Japanese personas link it to social friction and failing to meet obligations (, *meiwaku*).

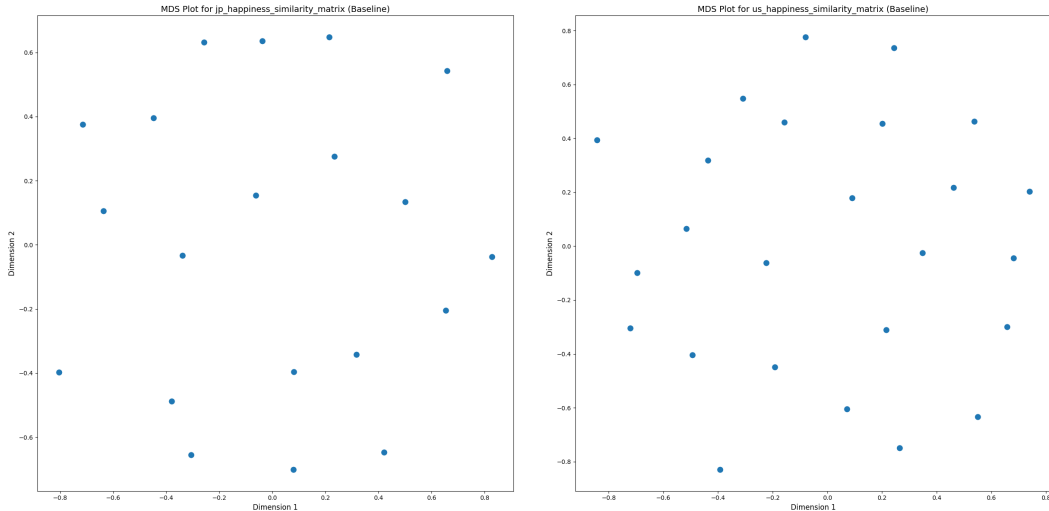


Figure 1: MDS plots visualizing the conceptual structure of happiness for Japanese (left) and US (right) personas in the baseline condition. The Japanese plot shows clusters related to social harmony and interpersonal relationships, while the US plot clusters around themes of personal achievement and autonomy.

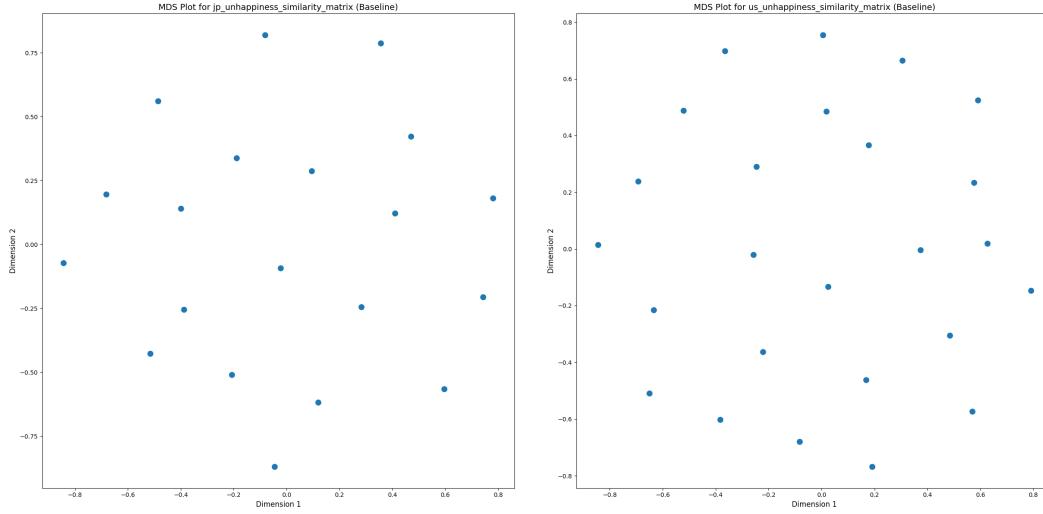


Figure 2: MDS plots visualizing the conceptual structure of unhappiness for Japanese (left) and US (right) personas in the baseline condition. The Japanese plot reveals themes of social friction and failure to meet obligations, while the US plot centers on personal failure and loss of freedom.

6.2 IMPACT OF THE CULTURALLY-AWARE PROMPT

Our most significant finding comes from the condition using a “culturally-aware” system prompt. As shown in Figure 3 and Figure 4, this prompt, which explicitly instructed the LLM to consider cultural context, produced a stark qualitative improvement. The responses became more nuanced, and the thematic clusters in the corresponding MDS plots grew tighter and more distinct. US personas more strongly emphasized self-oriented success, while Japanese personas focused more acutely on group harmony. This demonstrates that explicit contextual priming is an effective method for enhancing the cultural alignment of LLM responses.

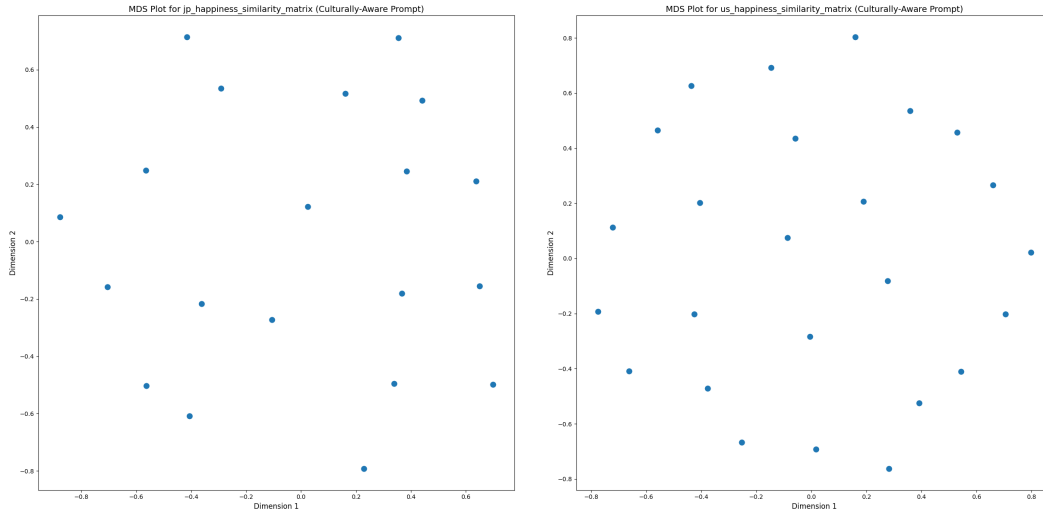


Figure 3: MDS plots for happiness under the culturally-aware prompt. The thematic clusters are more distinct than the baseline, with Japanese responses (left) more tightly focused on social harmony and US responses (right) more clearly emphasizing personal achievement.

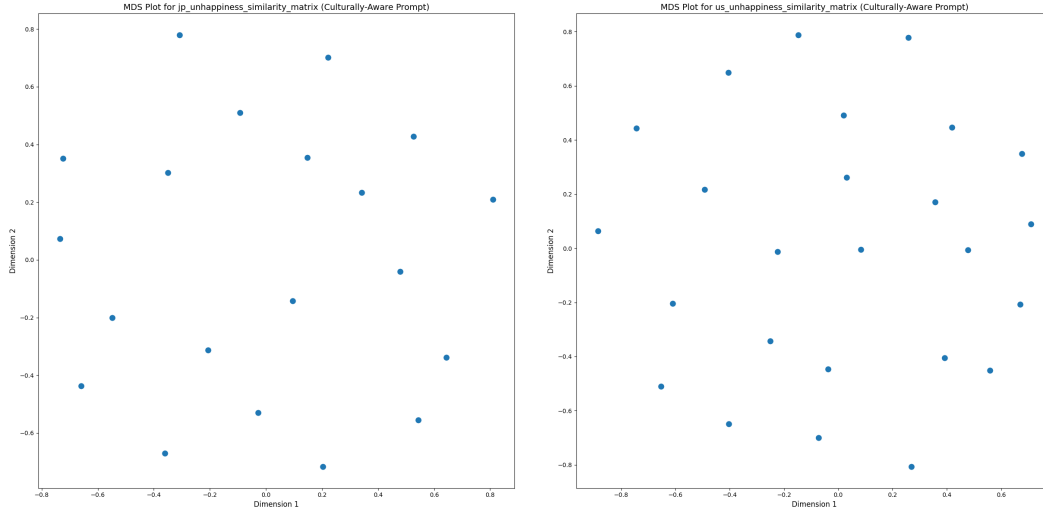


Figure 4: MDS plots for unhappiness under the culturally-aware prompt. The cultural distinctions are sharper, with Japanese personas (left) more clearly articulating unhappiness from social friction and US personas (right) focusing on thwarted personal ambition.

6.3 LIMITATIONS

We acknowledge several limitations. Our findings are based on a small number of personas and constitute a qualitative exploration rather than a quantitative measurement. The simulation, while controlled, cannot capture the full complexity of human experience. Future work should expand the diversity of personas and explore quantitative metrics for cultural alignment.

7 CONCLUSIONS AND FUTURE WORK

We introduced a methodology to evaluate LLM cultural alignment by simulating a cross-cultural psychology study with detailed US and Japanese personas. Our findings show that LLMs can produce culturally-consistent themes: US personas associate happiness with personal achievement, while Japanese personas link it to social harmony. Crucially, we demonstrated that a “culturally-aware” system prompt significantly sharpens these distinctions, offering a practical technique for eliciting deeper cultural alignment.

Building on this work, future research should extend this methodology to more diverse cultures and personas to test its generalizability. The development of quantitative metrics is a key next step for enabling more rigorous measurement of cultural alignment. This simulation approach could also be applied to other culturally-dependent concepts, such as ethical reasoning. Finally, investigating how model architecture and training data influence these cultural representations is essential for building more globally-aware and equitable AI systems.

This work was generated by THE AI SCIENTIST (Lu et al., 2024).

REFERENCES

Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The AI Scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*, 2024.